

Predicting salary from position level

A clear baseline for how much hierarchy level explains annual compensation.

The project asks a simple question: how much does level explain salary?

GOAL

Model annual compensation

Use job hierarchy level with Simple Linear Regression.

TOOLS

Build and inspect

Python, pandas, scikit-learn, and matplotlib.

FRAMING

Start with a baseline

A clear baseline before adding richer organizational features.

The dataset turns job titles into one usable feature

Position stays categorical; **Level (X)** becomes the numerical input; **Salary (y)** is annual compensation in USD.

POSITION	LEVEL	SALARY
Business Analyst	1	\$45,000
Junior C	2	\$50,000
Senior C	3	\$55,000
Manager	4	\$65,000
Contry Manager	5	\$75,000

Position becomes context; Level becomes X; Salary becomes y.

The Python pipeline moves from raw file to evaluated model

1. Import plotting, numerical, dataframe, regression, and metric libraries
2. Load Position_Salary_Data.csv and isolate $X = \text{Level}$, $y = \text{Salary}$
3. Fit LinearRegression, predict y_{pred} , then calculate R^2 and RMSE

PIPELINE FRAGMENT

```
X = Level  
y = Salary  
LinearRegression → y_pred  
R2 + RMSE
```

The fitted line says each level step is worth about \$49.7k

SLOPE

49,696.97 average salary increase per one-level step

Intercept: -94,333.33

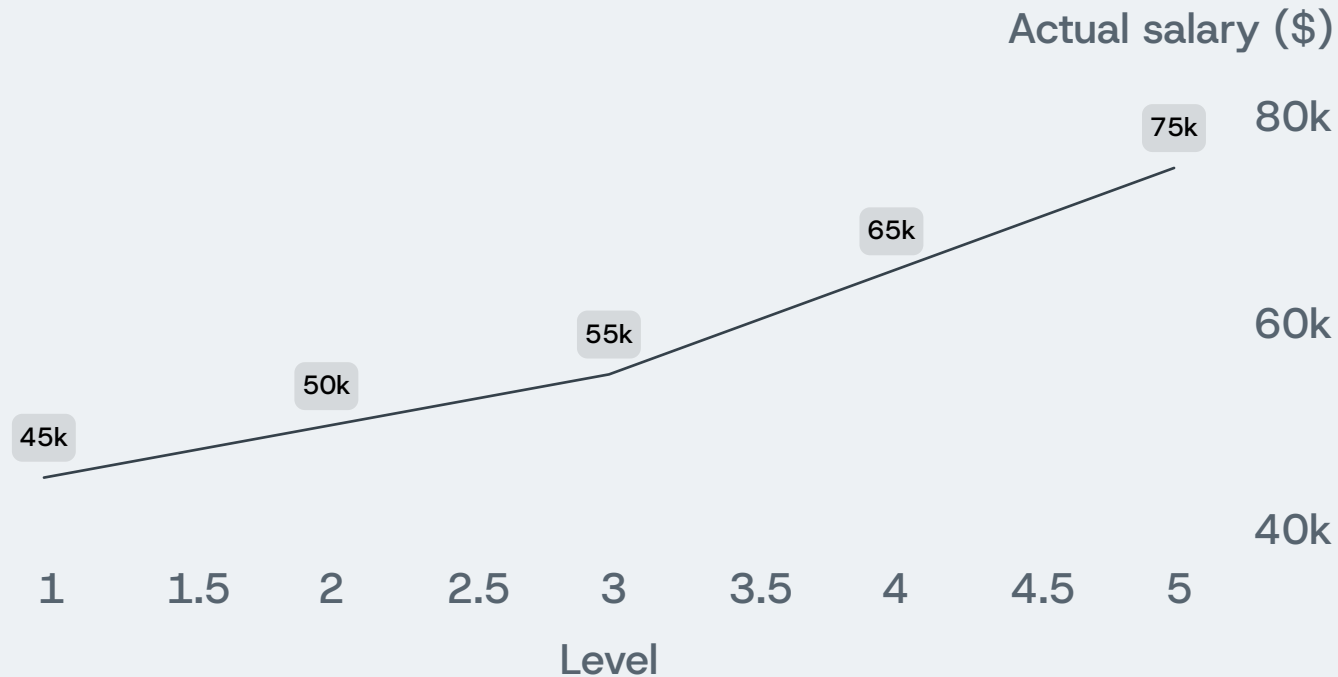
Interpretation: the model captures a strong upward relationship, but the intercept is an extrapolated baseline rather than a real job level.

SALARY BY POSITION LEVEL

LEVEL	ACTUAL SALARY
1	\$45,000
2	\$50,000
3	\$55,000
4	\$65,000
5	\$75,000

Salary rises with position level, with the fitted line summarizing the overall upward trend.

The fitted line says each level step is worth about \$49.7k



COEFFICIENT READOUT

\$49.7k

Slope: 49,696.97 average salary increase per one-level step.

Intercept: -94,333.33.

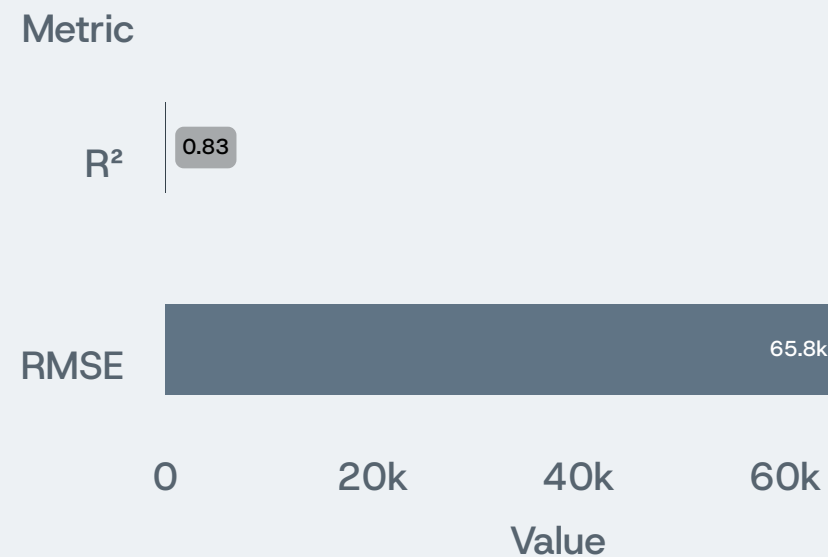
The intercept is an extrapolated baseline, not a real job level.

$R^2 = 0.825$: level explains most of the observed variation

0.825

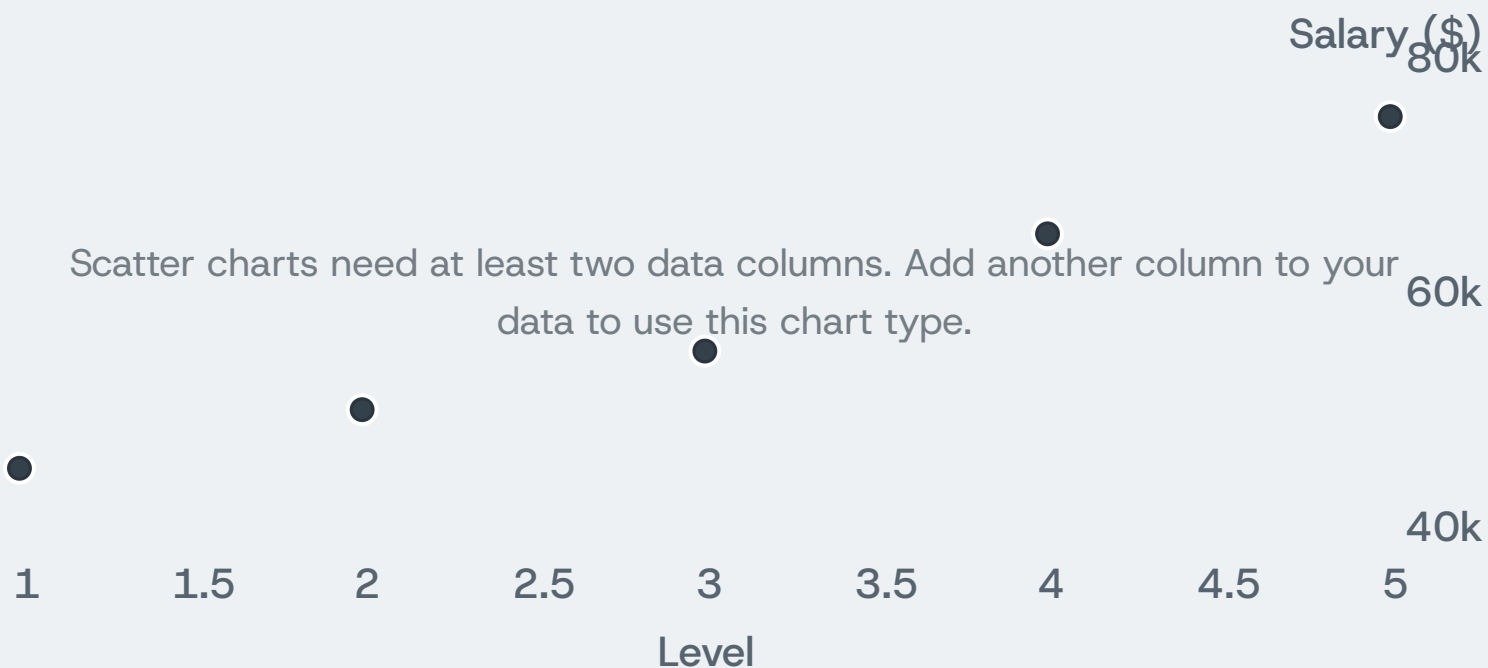
R^2 · approximately 82.5% of salary variance is explained linearly by position level.

RMSE: \$65,751.37, the average magnitude of prediction error across all levels. Notebook output reported for 10 data points.



The scatter plot makes the model's strengths and gaps visible

● Salary (\$)



Red points · actual salary observations.

Blue line · best-fit prediction path found by minimizing squared residuals.

Look for the upward direction, distance from the line, and where the linear assumption begins to strain.

Two upgrades can turn the baseline into a better compensation model

01

Polynomial Regression

Add X^2 and higher-order terms to capture nonlinear wage acceleration at executive levels.

02

Feature expansion

Add years of experience, department, and region for Multiple Linear Regression.

03

Compare models

Keep R^2 and RMSE as the common scorecard for the next iteration.

Simple linear regression is a strong first baseline, not the final answer.